Fatio, Rationally Reconstructed: A Faithful Deep and Shallow Embedding of a Multi-Agent Argumentation Protocol

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Abstract

We present a rational reconstruction of the LogiKEy implementation of the Fatio dialogue protocol for multi-agent dialectical argumentation (Pasetto and Benzmüller, ARQNL 2024), rebuilt following the methodology of "Faithful Logic Embeddings in HOL — Deep and Shallow" (Benzmüller 2025; AFP entry FaithfulPMLinHOL). The object logic of Fatio's axiomatic semantics — a multi-agent belief/desire logic over a propositional content logic, enriched with Done-, entailment- and existential-justification atoms — is developed both as a deep embedding (datatype syntax with a primitive-recursive Kripke semantics over models with explicit world domains) and as two shallow embeddings — a *maximal* one (model-parametric truth-set functions) and a *minimal* one in the classical LogiKEy style (fixed meta-level constants, truth sets over worlds) — all connected by machine-checked faithfulness theorems, including a soundness characterisation relating the two shallow backends and a documented comparison of their respective merits. The protocol layer (dialectical obligation stores, pre- and post-conditions, recursive dialogue checking) is defined over the deep syntax, so that syntactically distinct but semantically equivalent contents are discriminated exactly where the protocol requires it; the discrimination and identification facts, as well as the quantifier-scope question raised in the ARQNL paper, become theorems of the reconstruction.

Contents

1 Preliminaries

theory *FatioFaithful-preliminaries*
imports *Main*
begin

typedecl w — type of possible worlds
typedecl $\mathcal{S}$ — type of propositional constant symbols of the content logic
consts p::$\mathcal{S}$ q::$\mathcal{S}$ r::$\mathcal{S}$ n::$\mathcal{S}$ m::$\mathcal{S}$ s::$\mathcal{S}$ — some content atoms (as in the Fatio paper)
datatype *Speaker* = a | b | c — the dialogue participants

type-synonym $\mathcal{W}$ = w⇒*bool* — world domains
type-synonym $\mathcal{R}$ = *Speaker*⇒w⇒w⇒*bool* — agent-indexed accessibility relations
type-synonym $\mathcal{V}$ = $\mathcal{S}$⇒w⇒*bool* — valuations of content atoms

— Bounded universal quantifier, as in [2]
abbreviation(*input*) *BoundedAll*::$\mathcal{W}$⇒$\mathcal{W}$⇒*bool* **where** *BoundedAll* W $\varphi \equiv \forall x.$ $W\ x \longrightarrow \varphi\ x$
syntax -*BoundedAll*:: *pttrn*⇒$\mathcal{W}$⇒*bool*⇒*bool* ((3∀(-/:-)./ -) [0, 0, 10] 10)
translations $\forall x{:}W.\ \varphi \rightleftharpoons$ *CONST BoundedAll* W $(\lambda x.\ \varphi)$

declare[[*syntax-ambiguity-warning*=*false*]]

end

2 Deep embedding: content logic, Fatio language, BD logic

theory *FatioFaithful-deep*
imports *FatioFaithful-preliminaries*
begin

2.1 Content logic (deeply embedded, propositional)

datatype *Formula* = *AtmC* $\mathcal{S}$ (-^{c} [*1000*] *999*) | *NegC Formula* (¬^{c}- [*96*] *96*)
| *ImpC Formula Formula* (**infixr** $\supset^c$ *93*)
definition *AndC* (**infixr** $\wedge^c$ *95*) **where** $\varphi \wedge^c \psi \equiv \neg^c(\varphi \supset^c \neg^c\psi)$
definition *OrC* (**infixr** $\vee^c$ *92*) **where** $\varphi \vee^c \psi \equiv \neg^c\varphi \supset^c \psi$
definition *TopC* ($\top^c$) **where** $\top^c \equiv p^c \supset^c p^c$
definition *BotC* ($\bot^c$) **where** $\bot^c \equiv \neg^c\top^c$

2.2 The Fatio language

datatype *Sign* = *Pls* (⊕) | *Mns* (⊖)
datatype *FatioL* =
Assert Speaker Formula (*assert*[-,-])
| *Question Speaker Speaker Formula* (*question*[-,-,-])
| *Challenge Speaker Speaker Formula* (*challenge*[-,-,-])
| *Justify Speaker Formula Sign Formula* (*justify*[-,-⊢$^{-}$-])
| *Retract Speaker Formula Sign* (*retract*[-,-,-])

2.3 BD logic (deeply embedded)

The object language of the axiomatic semantics of Fatio [6], as formalised in [7]: agent beliefs $\mathcal{B}^d$ and desires $\mathcal{D}^d$ over the content logic, with three further kinds of atoms: *Done*^{d} for uttered locutions, *EntD* for argumentative entailment, and *ExJD i* φ whose single semantic clause carries the existential of the pre-conditions of question and challenge, $(\exists\,\Delta)$ *Done*[*justify*(*i*,$\Delta\vdash^{sg}\varphi$)], inside the object language. Following the sign-parametric justify locution of the LogiKEy sources [3], *ExJD* carries the sign of the requested justification.

datatype *BDF* =
AtmD $\mathcal{S}$ (-^{a} [*1000*] *999*)
| *DoneD FatioL* (*Done*^{d})
| *EntD Formula Sign Formula*
| *ExJD Speaker Sign Formula*
| *NegD BDF* (¬^{d}- [*96*] *96*)
| *ImpD BDF BDF* (**infixr** $\supset^d$ *93*)
| *BelD Speaker BDF* ($\mathcal{B}^d$)
| *DesD Speaker BDF* ($\mathcal{D}^d$)

definition *AndD* (**infixr** $\wedge^d$ *95*) **where** $\varphi \wedge^d \psi \equiv \neg^d(\varphi \supset^d \neg^d\psi)$
definition *OrD* (**infixr** $\vee^d$ *92*) **where** $\varphi \vee^d \psi \equiv \neg^d\varphi \supset^d \psi$

— Homomorphic injection of content formulas into the BD language
primrec *MapC* :: *Formula*⇒*BDF* ({-}^{d}) **where**
$\{\varphi^c\}^d = \varphi^a \mid \{\neg^c\varphi\}^d = \neg^d\{\varphi\}^d \mid \{\varphi \supset^c \psi\}^d = \{\varphi\}^d \supset^d \{\psi\}^d$

2.4 Kripke semantics over models $\langle W,B,D,V,U,E\rangle$

type-synonym $\mathcal{U}$ = *FatioL*⇒w⇒*bool* — Done-valuations
type-synonym $\mathcal{E}$ = *Formula*⇒*Sign*⇒*Formula*⇒w⇒*bool* — entailment-valuations

primrec *RelTD* :: $\mathcal{W}$⇒$\mathcal{R}$⇒$\mathcal{R}$⇒$\mathcal{V}$⇒$\mathcal{U}$⇒$\mathcal{E}$⇒w⇒*BDF*⇒*bool* (⟨-,-,-,-,-,-⟩,- $\models^d$ -) **where**
$\langle W,B,D,V,U,E\rangle,w \models^d x^a = V\ x\ w$
$\mid \langle W,B,D,V,U,E\rangle,w \models^d \mathit{Done}^d\ l = U\ l\ w$
$\mid \langle W,B,D,V,U,E\rangle,w \models^d \mathit{EntD}\ \Phi\ sg\ \varphi = E\ \Phi\ sg\ \varphi\ w$
$\mid \langle W,B,D,V,U,E\rangle,w \models^d \mathit{ExJD}\ i\ sg\ \varphi = (\exists\Delta.\ U\ (\mathit{Justify}\ i\ \Delta\ sg\ \varphi)\ w)$
$\mid \langle W,B,D,V,U,E\rangle,w \models^d \neg^d\varphi = (\neg\ \langle W,B,D,V,U,E\rangle,w \models^d \varphi)$
$\mid \langle W,B,D,V,U,E\rangle,w \models^d \varphi \supset^d \psi = (\langle W,B,D,V,U,E\rangle,w \models^d \varphi \longrightarrow \langle W,B,D,V,U,E\rangle,w \models^d \psi)$
$\mid \langle W,B,D,V,U,E\rangle,w \models^d \mathcal{B}^d\ i\ \varphi = (\forall v{:}W.\ B\ i\ w\ v \longrightarrow \langle W,B,D,V,U,E\rangle,v \models^d \varphi)$
$\mid \langle W,B,D,V,U,E\rangle,w \models^d \mathcal{D}^d\ i\ \varphi = (\forall v{:}W.\ D\ i\ w\ v \longrightarrow \langle W,B,D,V,U,E\rangle,v \models^d \varphi)$

definition *ValD* ($\models^d$ -) **where** $\models^d \varphi \equiv \forall\ W\ B\ D\ V\ U\ E.\ \forall w{:}W.\ \langle W,B,D,V,U,E\rangle,w \models^d \varphi$

— Global consequence from a set of deep premises, over all models
definition *ConsD* :: (*BDF*⇒*bool*)⇒*BDF*⇒*bool* (**infix** $\models$ *25*) **where**
$\Gamma \models \varphi \equiv \forall\ W\ B\ D\ V\ U\ E.\ (\forall\gamma.\ \Gamma\ \gamma \longrightarrow (\forall w{:}W.\ \langle W,B,D,V,U,E\rangle,w \models^d \gamma)) \longrightarrow (\forall w{:}W.\ \langle W,B,D,V,U,E\rangle,w \models^d \varphi)$

named-theorems *DefD*
declare *AndC-def*[*DefD*,*simp*] *OrC-def*[*DefD*,*simp*] *TopC-def*[*DefD*,*simp*] *BotC-def*[*DefD*,*simp*] *AndD-def*[*DefD*,*simp*] *OrD-def*[*DefD*,*simp*] *ValD-def*[*DefD*] *ConsD-def*[*DefD*]

end

3 Shallow embedding of the BD logic (maximal)

theory *FatioFaithful-shallow*
imports *FatioFaithful-deep*
begin

type-synonym σ = $\mathcal{W}$⇒$\mathcal{R}$⇒$\mathcal{R}$⇒$\mathcal{V}$⇒$\mathcal{U}$⇒$\mathcal{E}$⇒w⇒*bool*

definition *AtmS*::$\mathcal{S}$⇒σ (-^{s} [*1000*] *999*) **where** $x^s \equiv \lambda W\ B\ D\ V\ U\ E\ w.\ V\ x\ w$
definition *DoneS*::*FatioL*⇒σ (*Done*^{s}) **where** $\mathit{Done}^s\ l \equiv \lambda W\ B\ D\ V\ U\ E\ w.\ U\ l\ w$
definition *EntS*::*Formula*⇒*Sign*⇒*Formula*⇒σ **where** $\mathit{EntS}\ \Phi\ sg\ \varphi \equiv \lambda W\ B\ D\ V\ U\ E\ w.\ E\ \Phi\ sg\ \varphi\ w$

definition *ExJS*::*Speaker*⇒*Sign*⇒*Formula*⇒σ **where**
ExJS i sg $\varphi \equiv \lambda W\ B\ D\ V\ U\ E\ w.\ \exists \Delta.\ U$ (*Justify i* Δ *sg* φ) w
definition *NegS*::σ⇒σ ($\neg^s$- [*96*] *96*) **where** $\neg^s\varphi \equiv \lambda W\ B\ D\ V\ U\ E\ w.\ \neg\ \varphi\ W\ B\ D\ V\ U\ E\ w$
definition *ImpS*::σ⇒σ⇒σ (**infixr** $\supset^s$ *93*) **where**
$\varphi \supset^s \psi \equiv \lambda W\ B\ D\ V\ U\ E\ w.\ \varphi\ W\ B\ D\ V\ U\ E\ w \longrightarrow \psi\ W\ B\ D\ V\ U\ E\ w$
definition *BelS*::*Speaker*⇒σ⇒σ ($\mathcal{B}^s$) **where**
$\mathcal{B}^s\ i\ \varphi \equiv \lambda W\ B\ D\ V\ U\ E\ w.\ \forall v{:}W.\ B\ i\ w\ v \longrightarrow \varphi\ W\ B\ D\ V\ U\ E\ v$
definition *DesS*::*Speaker*⇒σ⇒σ ($\mathcal{D}^s$) **where**
$\mathcal{D}^s\ i\ \varphi \equiv \lambda W\ B\ D\ V\ U\ E\ w.\ \forall v{:}W.\ D\ i\ w\ v \longrightarrow \varphi\ W\ B\ D\ V\ U\ E\ v$

definition *RelTS*::$\mathcal{W}$⇒$\mathcal{R}$⇒$\mathcal{R}$⇒$\mathcal{V}$⇒$\mathcal{U}$⇒$\mathcal{E}$⇒w⇒σ⇒*bool* ($\langle$-,-,-,-,-,-$\rangle$,- $\models^s$ -) **where**
$\langle W,B,D,V,U,E\rangle,w \models^s \varphi \equiv \varphi\ W\ B\ D\ V\ U\ E\ w$
definition *ValS* ($\models^s$ -) **where** $\models^s \varphi \equiv \forall\ W\ B\ D\ V\ U\ E.\ \forall\ w{:}W.\ \langle W,B,D,V,U,E\rangle,w \models^s \varphi$

named-theorems *DefS*
declare *AtmS-def*[*DefS*,*simp*] *DoneS-def*[*DefS*,*simp*] *EntS-def*[*DefS*,*simp*] *ExJS-def*[*DefS*,*simp*] *NegS-def*[*DefS*,*simp*] *ImpS-def*[*DefS*,*simp*] *BelS-def*[*DefS*,*simp*] *DesS-def*[*DefS*,*simp*] *RelTS-def*[*DefS*,*simp*] *ValS-def*[*DefS*]

end

4 Shallow embedding of the BD logic (minimal)

theory *FatioFaithful-shallow-minimal*
imports *FatioFaithful-deep*
begin

The minimal shallow embedding fixes one model at the meta level: the accessibility relations and valuations are uninterpreted HOL constants, formulas are plain truth sets over worlds, and the world domain is implicitly the full type w. This is the classical SSE style of LogiKEy [3] and the style of the original HOMML/FATIO development; the price, made explicit by the faithfulness theorems below, is that faithfulness holds relative to full-domain models only.

consts B_0::$\mathcal{R}$ D_0::$\mathcal{R}$ V_0::$\mathcal{V}$ U_0::$\mathcal{U}$ E_0::$\mathcal{E}$
consts V_1::$\mathcal{V}$ — a second valuation, used for a countermodel below

type-synonym σ_m = w⇒*bool*

definition *AtmM*::$\mathcal{S}$⇒σ_m ($^{-m}$ [*1000*] *999*) **where** $x^m \equiv \lambda w.\ V_0\ x\ w$
definition *DoneM*::*FatioL*⇒σ_m (*Done*^{m}) **where** *Done*^{m} $l \equiv \lambda w.\ U_0\ l\ w$
definition *EntM*::*Formula*⇒*Sign*⇒*Formula*⇒σ_m **where** *EntM* Φ *sg* $\varphi \equiv \lambda w.\ E_0\ \Phi$ *sg* φ w
definition *ExJM*::*Speaker*⇒*Sign*⇒*Formula*⇒σ_m **where**
ExJM i sg $\varphi \equiv \lambda w.\ \exists \Delta.\ U_0$ (*Justify i* Δ *sg* φ) w
definition *NegM*::σ_m⇒σ_m ($\neg^m$- [*96*] *96*) **where** $\neg^m\varphi \equiv \lambda w.\ \neg\ \varphi\ w$

definition *ImpM*::$\sigma_m \Rightarrow \sigma_m \Rightarrow \sigma_m$ (**infixr** $\supset^m$ *93*) **where** $\varphi \supset^m \psi \equiv \lambda w.\ \varphi\ w \longrightarrow \psi\ w$
definition *BelM*::*Speaker*$\Rightarrow\sigma_m\Rightarrow\sigma_m$ ($\mathcal{B}^m$) **where** $\mathcal{B}^m\ i\ \varphi \equiv \lambda w.\ \forall\, v.\ B_0\ i\ w\ v \longrightarrow \varphi\ v$
definition *DesM*::*Speaker*$\Rightarrow\sigma_m\Rightarrow\sigma_m$ ($\mathcal{D}^m$) **where** $\mathcal{D}^m\ i\ \varphi \equiv \lambda w.\ \forall\, v.\ D_0\ i\ w\ v \longrightarrow \varphi\ v$

definition *RelTM*::w$\Rightarrow\sigma_m\Rightarrow$*bool* (- $\models^m$ -) **where** $w \models^m \varphi \equiv \varphi\ w$
definition *ValM* ($\models^m$ -) **where** $\models^m \varphi \equiv \forall\, w.\ \varphi\ w$

named-theorems *DefM*
declare *AtmM-def*[*DefM*,*simp*] *DoneM-def*[*DefM*,*simp*] *EntM-def*[*DefM*,*simp*] *ExJM-def*[*DefM*,*simp*] *NegM-def*[*DefM*,*simp*] *ImpM-def*[*DefM*,*simp*] *BelM-def*[*DefM*,*simp*] *DesM-def*[*DefM*,*simp*] *RelTM-def*[*DefM*,*simp*] *ValM-def*[*DefM*]

end

5 Faithfulness

theory *FatioFaithful-faithfulness*
imports *FatioFaithful-shallow FatioFaithful-shallow-minimal*
begin

— Mapping: deep to (maximal) shallow
primrec *DpToSh* :: *BDF*$\Rightarrow\sigma$ ($(\!|$-$|\!)$) **where**
$(\!|x^a|\!) = x^s$ | $(\!|Done^d\ l|\!) = Done^s\ l$ | $(\!|EntD\ \Phi\ sg\ \varphi|\!) = EntS\ \Phi\ sg\ \varphi$
| $(\!|ExJD\ i\ sg\ \varphi|\!) = ExJS\ i\ sg\ \varphi$ | $(\!|\neg^d\varphi|\!) = \neg^s(\!|\varphi|\!)$ | $(\!|\varphi \supset^d \psi|\!) = (\!|\varphi|\!) \supset^s (\!|\psi|\!)$
| $(\!|\mathcal{B}^d\ i\ \varphi|\!) = \mathcal{B}^s\ i\ (\!|\varphi|\!)$ | $(\!|\mathcal{D}^d\ i\ \varphi|\!) = \mathcal{D}^s\ i\ (\!|\varphi|\!)$

— Automated faithfulness proofs, as in [1,2]
theorem *Faithful1a*:
$\forall\, W\ B\ D\ V\ U\ E.\ \forall\, w{:}W.\ \langle W,B,D,V,U,E\rangle,w \models^d \varphi \longleftrightarrow \langle W,B,D,V,U,E\rangle,w \models^s (\!|\varphi|\!)$
apply (*induct* φ) **by** *auto*

theorem *Faithful1b*: $\models^d \varphi \longleftrightarrow \models^s (\!|\varphi|\!)$
using *Faithful1a* **unfolding** *ValD-def ValS-def* **by** *auto*

5.1 Discrimination vs. identification: the Done-problem, resolved

Syntactically, $p \supset^c p$ and $\neg^c\bot^c$ are different contents; hence locutions and Done-atoms over them are different objects. Semantically, their injections into the BD logic are equivalent. All four facts are theorems of the reconstruction.

lemma *ContentDiscriminated*: $(p^c \supset^c p^c) \neq \neg^c\bot^c$ **by** *simp*
lemma *LocutionDiscriminated*: *assert*[a, $p^c \supset^c p^c$] $\neq$ *assert*[a, $\neg^c\bot^c$] **by** *simp*

lemma *DoneDiscriminated*: $Done^d\ assert[a,\ p^c\ \supset^c\ p^c] \neq Done^d\ assert[a,\ \neg^c\bot^c]$
by *simp*
lemma *SemanticsIdentifies*:
$\models^d (\{p^c\ \supset^c\ p^c\}^d\ \supset^d\ \{\neg^c\bot^c\}^d) \wedge \models^d (\{\neg^c\bot^c\}^d\ \supset^d\ \{p^c\ \supset^c\ p^c\}^d)$
unfolding *ValD-def* **by** *simp*

5.2 The quantifier-scope question, settled

The wide-scope (meta-level) existential of the pre-conditions of question and challenge implies the narrow-scope (object-level, via *ExJD*) reading.

lemma *ScopeImp*:
assumes $\exists\,\Delta.\ \Gamma \models \mathcal{D}^d\ j\ (\mathcal{B}^d\ k\ (\mathcal{D}^d\ j\ (Done^d\ (Justify\ i\ \Delta\ sg\ \varphi))))$
shows $\Gamma \models \mathcal{D}^d\ j\ (\mathcal{B}^d\ k\ (\mathcal{D}^d\ j\ (ExJD\ i\ sg\ \varphi)))$
proof −
obtain Δ **where** H: $\Gamma \models \mathcal{D}^d\ j\ (\mathcal{B}^d\ k\ (\mathcal{D}^d\ j\ (Done^d\ (Justify\ i\ \Delta\ sg\ \varphi))))$ **using** *assms* **by** *blast*
show *?thesis* **unfolding** *ConsD-def*
proof (*intro allI impI*)
fix $W\ B\ D\ V\ U\ E\ w$
assume *sat*: $\forall\,\gamma.\ \Gamma\ \gamma \longrightarrow (\forall\,w{:}W.\ \langle W,B,D,V,U,E\rangle,w \models^d \gamma)$ **and** wW: $W\ w$
have $\langle W,B,D,V,U,E\rangle,w \models^d \mathcal{D}^d\ j\ (\mathcal{B}^d\ k\ (\mathcal{D}^d\ j\ (Done^d\ (Justify\ i\ \Delta\ sg\ \varphi))))$
using H[*unfolded ConsD-def*] *sat* wW **by** *blast*
thus $\langle W,B,D,V,U,E\rangle,w \models^d \mathcal{D}^d\ j\ (\mathcal{B}^d\ k\ (\mathcal{D}^d\ j\ (ExJD\ i\ sg\ \varphi)))$ **by** *simp blast*
qed
qed

The converse fails: the narrow-scope reading does NOT imply the wide-scope one. This is where a model finder earns its keep - the statement is a non-theorem, and *nitpick* [5] refutes it in seconds, from which the countermodel below was read off. Two worlds suffice: at each of them a justification is uttered, but for DIFFERENT supports, so no single Δ works globally. We verify the countermodel explicitly, so that the entry stays independent of the model finder at build time.

lemma *ScopeConverse-countermodel*:
fixes $\varphi_1\ \varphi_2$ :: *Formula* **and** $w_1\ w_2$ :: w **and** *i*::*Speaker* **and** *sg*::*Sign* **and** φ::*Formula*
assumes *distinct*Δ: $\varphi_1 \neq \varphi_2$ **and** *distinctw*: $w_1 \neq w_2$
and W: $W = (\lambda w.\ w = w_1 \vee w = w_2)$
and U: $U = (\lambda l\ w.\ l = Justify\ i\ (if\ w = w_1\ then\ \varphi_1\ else\ \varphi_2)\ sg\ \varphi)$
shows $(\forall\,w{:}W.\ \langle W,B,D,V,U,E\rangle,w \models^d ExJD\ i\ sg\ \varphi)$
$\wedge\ \neg(\exists\,\Delta.\ \forall\,w{:}W.\ \langle W,B,D,V,U,E\rangle,w \models^d Done^d\ (Justify\ i\ \Delta\ sg\ \varphi))$
proof
show $\forall\,w{:}W.\ \langle W,B,D,V,U,E\rangle,w \models^d ExJD\ i\ sg\ \varphi$ **using** U **by** *auto*
next
show $\neg(\exists\,\Delta.\ \forall\,w{:}W.\ \langle W,B,D,V,U,E\rangle,w \models^d Done^d\ (Justify\ i\ \Delta\ sg\ \varphi))$
proof
assume $\exists\,\Delta.\ \forall\,w{:}W.\ \langle W,B,D,V,U,E\rangle,w \models^d Done^d\ (Justify\ i\ \Delta\ sg\ \varphi)$
then obtain Δ **where** $U\ (Justify\ i\ \Delta\ sg\ \varphi)\ w_1$ **and** $U\ (Justify\ i\ \Delta\ sg\ \varphi)\ w_2$

using *W* **by** *auto*
hence $\Delta = \varphi_1$ **and** $\Delta = \varphi_2$ **using** *U distinctw* **by** *auto*
thus *False* **using** *distinct*Δ **by** *simp*
qed
qed

5.3 Faithfulness for the minimal embedding, and the comparison

primrec *DpToShMin* :: *BDF*$\Rightarrow\sigma_m$ ($\lceil$-$\rceil$) **where**
$\lceil x^a \rceil = x^m$ | $\lceil Done^d\ l \rceil = Done^m\ l$ | $\lceil EntD\ \Phi\ sg\ \varphi \rceil = EntM\ \Phi\ sg\ \varphi$
| $\lceil ExJD\ i\ sg\ \varphi \rceil = ExJM\ i\ sg\ \varphi$ | $\lceil \neg^d \varphi \rceil = \neg^m \lceil \varphi \rceil$ | $\lceil \varphi \supset^d \psi \rceil = \lceil \varphi \rceil \supset^m \lceil \psi \rceil$
| $\lceil \mathcal{B}^d\ i\ \varphi \rceil = \mathcal{B}^m\ i\ \lceil \varphi \rceil$ | $\lceil \mathcal{D}^d\ i\ \varphi \rceil = \mathcal{D}^m\ i\ \lceil \varphi \rceil$

Faithfulness for the minimal embedding holds relative to the one fixed, full-domain model given by the meta-level constants.

theorem *Faithful2*:
$\forall\, w.\ \langle(\lambda x{::}\text{w.}\ True), B_0, D_0, V_0, U_0, E_0\rangle, w \models^d \varphi \longleftrightarrow w \models^m \lceil \varphi \rceil$
apply (*induct* φ) **by** *auto*

theorem *Faithful3*:
$\forall\, w.\ \langle(\lambda x{::}\text{w.}\ True), B_0, D_0, V_0, U_0, E_0\rangle, w \models^s (\!|\varphi|\!) \longleftrightarrow w \models^m \lceil \varphi \rceil$
using *Faithful1a*[*of* φ] *Faithful2*[*of* φ] **by** *auto*

Minimal validity is sound for deep validity in the following precise sense: the minimally valid truth sets are exactly the images of the deeply valid formulas. (This ports lemma Sound1 of FaithfulPMLinHOL, with a structured proof.)

theorem *SoundMin*: $(\models^m \psi) \longleftrightarrow (\exists\, \varphi.\ \psi = \lceil \varphi \rceil \wedge \models^d \varphi)$
proof
assume *L*: $\models^m \psi$
have $\psi = \lceil p^a \supset^d p^a \rceil$
proof (*rule ext*)
fix *w* **show** $\psi\ w = \lceil p^a \supset^d p^a \rceil\ w$ **using** *L* **unfolding** *ValM-def* **by** *simp*
qed
moreover have $\models^d (p^a \supset^d p^a)$ **unfolding** *ValD-def* **by** *simp*
ultimately show $\exists\, \varphi.\ \psi = \lceil \varphi \rceil \wedge \models^d \varphi$ **by** *blast*
next
assume $\exists\, \varphi.\ \psi = \lceil \varphi \rceil \wedge \models^d \varphi$
then obtain φ **where** *1*: $\psi = \lceil \varphi \rceil$ **and** *2*: $\models^d \varphi$ **by** *blast*
have $\forall\, w.\ \langle(\lambda x{::}\text{w.}\ True), B_0, D_0, V_0, U_0, E_0\rangle, w \models^d \varphi$ **using** *2* **unfolding** *ValD-def*
by *simp*
thus $\models^m \psi$ **using** *1 Faithful2* **unfolding** *ValM-def* **by** *simp*
qed

A second use of the model finder: the two shallow backends are NOT interchangeable. Minimal validity of $\lceil \varphi \rceil$ does not entail deep validity of φ, since the fixed constants may satisfy a formula that fails in some other model.

nitpick refutes the implication; the witness below makes the refutation permanent: an atom true throughout the fixed model but false somewhere else.

lemma *MinimalNotDeep-countermodel*:
  assumes *minimally-valid*: $\forall\, w.\ V_0\ x\ w$ **and** *elsewhere-false*: $\neg\ V_1\ x\ w_1$
  shows $\models^m\ \lceil x^a \rceil$ **and** $\neg\ \models^d\ x^a$
proof −
  show $\models^m\ \lceil x^a \rceil$ **using** *minimally-valid* **unfolding** *ValM-def* **by** *simp*
next
  show $\neg\ \models^d\ x^a$
  proof
    assume *A*: $\models^d\ x^a$
    have $V_1\ x\ w_1$
      using *A*[*unfolded ValD-def*,
            THEN spec[**where** $x{=}\lambda w{::}\mathrm{w}.\ \mathit{True}$], *THEN spec*[**where** $x{=}B_0$],
            THEN spec[**where** $x{=}D_0$], *THEN spec*[**where** $x{=}V_1$],
            THEN spec[**where** $x{=}U_0$], *THEN spec*[**where** $x{=}E_0$],
            THEN spec[**where** $x{=}w_1$]] **by** *simp*
    thus *False* **using** *elsewhere-false* **by** *simp*
  qed
qed

Comparison. The maximal embedding threads the whole model through every formula: validity and the global consequence relation used by the protocol are genuinely model-quantified, faithfulness (Faithful1a/1b) is unrestricted, and bounded world domains remain expressible. The minimal embedding is leaner - one world argument instead of seven parameters - and matches the original HOMML/FATIO development; but its faithfulness (Faithful2) is relative to the one fixed full-domain model, so dialogue checking on this backend is checking IN a situation rather than a logically necessary entailment. Since the protocol layer is defined over the DEEP syntax, both backends serve it soundly; this entry uses the maximal one for the protocol's consequence relation and provides the minimal one, with the bridging theorems above, for lightweight automation and for continuity with the original sources.

end

6 The Fatio protocol over the deep syntax

theory *FatioFaithful-protocol*
  imports *FatioFaithful-faithfulness*
begin

6.1 Dialectical obligation stores

datatype *DOSEntry* = *Entry Speaker Formula Sign*
type-synonym *DOS* = *DOSEntry list*

fun *DOSUpdate* :: *FatioL*⇒*DOS*⇒*DOS* **where**
  DOSUpdate assert[*i*,φ] *dos* = (*Entry i* φ $\oplus$) # *dos*
| *DOSUpdate question*[*j*,*i*,φ] *dos* = *dos*
| *DOSUpdate challenge*[*j*,*i*,φ] *dos* = (*Entry j* φ $\ominus$) # *dos*
| *DOSUpdate* (*Justify i* Φ *sg* φ) *dos* = (*Entry i* Φ *sg*) # *dos*
| *DOSUpdate retract*[*i*,φ,*sg*] *dos* = *removeAll* (*Entry i* φ *sg*) *dos*

6.2 Pre-conditions (axiomatic semantics)

Two reconstruction decisions are made explicit here. (1) Following the sources of [7], justify is sign-parametric: *justify*[i,$\Phi \vdash^{\ominus} \varphi$] discharges a negative obligation incurred by a challenge. (2) The pre-condition of question is generalised so that the requested justification matches the SIGN of the questioned speaker's actual obligation; the Brigade Restaurant dialogue (where c questions the challenger b, whose obligation is negative) requires exactly this generalisation, which remains tacit in the original sources.

fun *PreCond* :: *DOS*⇒(*BDF*⇒*bool*)⇒*FatioL*⇒*bool* **where**
  PreCond dos Γ *assert*[*i*,φ] =
    ((*Entry i* φ $\oplus$) $\notin$ *set dos* $\wedge$ ($\forall$ *j*. *j*≠*i* $\longrightarrow$ $\Gamma \models \mathcal{D}^d\ i\ (\mathcal{B}^d\ j\ (\mathcal{B}^d\ i\ \{\varphi\}^d))$))
| *PreCond dos* Γ *question*[*j*,*i*,φ] =
    (*i*≠*j* $\wedge$ ($\exists$ *sg*. (*Entry i* φ *sg*) $\in$ *set dos* $\wedge$
     ($\forall$ *k*. *k*≠*j* $\longrightarrow$ $\Gamma \models \mathcal{D}^d\ j\ (\mathcal{B}^d\ k\ (\mathcal{D}^d\ j\ (ExJD\ i\ sg\ \varphi))))$))
| *PreCond dos* Γ *challenge*[*j*,*i*,φ] =
    (*i*≠*j* $\wedge$ (*Entry i* φ $\oplus$) $\in$ *set dos* $\wedge$
     ($\forall$ *k*. *k*≠*j* $\longrightarrow$ $\Gamma \models \mathcal{D}^d\ j\ (\mathcal{B}^d\ k\ (\neg^d(\mathcal{B}^d\ j\ \{\varphi\}^d)))$) $\wedge$
     ($\forall$ *k*. *k*≠*j* $\longrightarrow$ $\Gamma \models \mathcal{D}^d\ j\ (\mathcal{B}^d\ k\ (\mathcal{D}^d\ j\ (ExJD\ i \oplus \varphi)))$))
| *PreCond dos* Γ (*Justify i* Φ *sg* φ) =
    ((*Entry i* φ *sg*) $\in$ *set dos* $\wedge$
     ($\exists$ *j*. *j*≠*i* $\wedge$ (Γ (*Done*^{d} *question*[*j*,*i*,φ]) $\vee$ Γ (*Done*^{d} *challenge*[*j*,*i*,φ])) $\wedge$
     ($\forall$ *k*. *k*≠*i* $\longrightarrow$ $\Gamma \models \mathcal{D}^d\ i\ (\mathcal{B}^d\ k\ (\mathcal{B}^d\ i\ (EntD\ \Phi\ sg\ \varphi)))$))
| *PreCond dos* Γ *retract*[*i*,φ,*sg*] =
    ((*Entry i* φ *sg*) $\in$ *set dos* $\wedge$
     (*case sg of* $\oplus$ ⇒ ($\forall$ *j*. *j*≠*i* $\longrightarrow$ $\Gamma \models \mathcal{D}^d\ i\ (\mathcal{B}^d\ j\ (\neg^d(\mathcal{B}^d\ i\ \{\varphi\}^d)))$)
       | $\ominus$ ⇒ ($\forall$ *j*. *j*≠*i* $\longrightarrow$ $\Gamma \models \mathcal{D}^d\ i\ (\mathcal{B}^d\ j\ (\neg^d\neg^d(\mathcal{B}^d\ i\ \{\varphi\}^d)))$)))

6.3 Post-conditions and state evolution

fun *GammaAdd* :: *FatioL*⇒(*BDF*⇒*bool*) **where**
  GammaAdd assert[*i*,φ] =
    ($\lambda x.\ \exists k\ j.\ k \neq i \wedge j \neq i \wedge x = \mathcal{B}^d\ k\ (\mathcal{D}^d\ i\ (\mathcal{B}^d\ j\ (\mathcal{B}^d\ i\ \{\varphi\}^d)))$)
| *GammaAdd question*[*j*,*i*,φ] = ($\lambda x.$ *False*)
| *GammaAdd challenge*[*j*,*i*,φ] = ($\lambda x.$ *False*)
| *GammaAdd* (*Justify i* Φ *sg* φ) =
    ($\lambda x.\ \exists k\ j.\ k \neq i \wedge j \neq i \wedge x = \mathcal{B}^d\ k\ (\mathcal{D}^d\ i\ (\mathcal{B}^d\ j\ (\mathcal{B}^d\ i\ (EntD\ \Phi\ sg\ \varphi))))$)
| *GammaAdd retract*[*i*,φ,*sg*] =
    (*case sg of* $\oplus$ ⇒ ($\lambda x.\ \exists k\ j.\ k \neq i \wedge j \neq i \wedge x = \mathcal{B}^d\ k\ (\mathcal{D}^d\ i\ (\mathcal{B}^d\ j\ (\neg^d(\mathcal{B}^d\ i\ \{\varphi\}^d)))))$
      | $\ominus$ ⇒ ($\lambda x.\ \exists k\ j.\ k \neq i \wedge j \neq i \wedge x = \mathcal{B}^d\ k\ (\mathcal{D}^d\ i\ (\mathcal{B}^d\ j\ (\neg^d\neg^d(\mathcal{B}^d\ i$

$\{\varphi\}^d)))))$

definition *GammaUpdate* :: *FatioL*⇒(*BDF*⇒*bool*)⇒(*BDF*⇒*bool*) **where**
GammaUpdate l Γ ≡ λ*x*. *GammaAdd l x* ∨ Γ *x* ∨ *x* = *Done*^{d} *l*

fun *FatioCheckRec* :: *FatioL list*⇒*DOS*⇒(*BDF*⇒*bool*)⇒*bool* **where**
FatioCheckRec [] *dos* Γ = *True*
| *FatioCheckRec* (*l*#*ls*) *dos* Γ =
(*PreCond dos* Γ *l* ∧ *FatioCheckRec ls* (*DOSUpdate l dos*) (*GammaUpdate l* Γ))

6.4 Sanity theorems

lemma *MemberEntails*[*intro*]: Γ φ ⟹ Γ ⊨ φ **unfolding** *ConsD-def* **by** *blast*
lemma *GammaMono*: Γ *x* ⟹ *GammaUpdate l* Γ *x* **by** (*simp add*: *GammaUpdate-def*)
lemma *DoneRecorded*: *GammaUpdate l* Γ (*Done*^{d} *l*) **by** (*simp add*: *GammaUpdate-def*)
lemma *AssertCreatesObligation*: (*Entry i* φ ⊕) ∈ *set* (*DOSUpdate assert*[*i*,φ] *dos*) **by** *simp*
lemma *RetractRemovesObligation*: (*Entry i* φ *sg*) ∉ *set* (*DOSUpdate retract*[*i*,φ,*sg*] *dos*) **by** *simp*

end

7 Tests: two dialogues, checked with high automation

theory *FatioFaithful-tests*
imports *FatioFaithful-protocol*
begin

7.1 The full Brigade Restaurant dialogue (from the literature)

The six-locution dialogue of [7], going back to [6]: a asserts that the Brigade restaurant is good (r); b challenges; a justifies from the newspaper review (n); c questions the challenger b, whose obligation is NEGATIVE; b justifies negatively from the chef change and quality drop ($m \wedge^c s$); finally a retracts. The initial knowledge Γ_B lists exactly the mental-state formulas required by the pre-conditions.

definition Γ_B :: *BDF*⇒*bool* **where** $\Gamma_B \equiv \lambda x.$
(∃ *j*. *j*≠*a* ∧ *x* = $\mathcal{D}^d$ *a* ($\mathcal{B}^d$ *j* ($\mathcal{B}^d$ *a* {r^c}^{d})))
∨ (∃ *k*. *k*≠*b* ∧ *x* = $\mathcal{D}^d$ *b* ($\mathcal{B}^d$ *k* (¬^{d}($\mathcal{B}^d$ *b* {r^c}^{d}))))
∨ (∃ *k*. *k*≠*b* ∧ *x* = $\mathcal{D}^d$ *b* ($\mathcal{B}^d$ *k* ($\mathcal{D}^d$ *b* (*ExJD a* ⊕ (r^c)))))
∨ (∃ *k*. *k*≠*a* ∧ *x* = $\mathcal{D}^d$ *a* ($\mathcal{B}^d$ *k* ($\mathcal{B}^d$ *a* (*EntD* (n^c) ⊕ (r^c)))))
∨ (∃ *k*. *k*≠*c* ∧ *x* = $\mathcal{D}^d$ *c* ($\mathcal{B}^d$ *k* ($\mathcal{D}^d$ *c* (*ExJD b* ⊖ (r^c)))))

$\vee\ (\exists\, k.\ k{\neq}b \wedge x = \mathcal{D}^d\ b\ (\mathcal{B}^d\ k\ (\mathcal{B}^d\ b\ (\mathit{EntD}\ (m^c \wedge^c s^c) \ominus (r^c)))))$
$\vee\ (\exists\, j.\ j{\neq}a \wedge x = \mathcal{D}^d\ a\ (\mathcal{B}^d\ j\ (\neg^d(\mathcal{B}^d\ a\ \{r^c\}^d))))$

abbreviation *Brigade* ≡
[*assert*[*a*,r^c], *challenge*[*b*,*a*,r^c], *justify*[*a*,$n^c \vdash^{\oplus} r^c$],
question[*c*,*b*,r^c], *justify*[*b*,$m^c \wedge^c s^c \vdash^{\ominus} r^c$], *retract*[*a*,$r^c$,⊕]]

Automation. Every step obligation is discharged by a single automated call (*auto* with the two protocol definitions and *MemberEntails* as an introduction rule). The justify steps additionally exhibit their Done-witness in two lines beforehand, and the question steps select the sign of the addressed obligation. This division of labour is deliberate: *sledgehammer* [4] finds no proof for these steps even with generous time limits and the relevant facts supplied, because what they require is the choice of a WITNESS over a finite datatype (which speaker questioned or challenged, which sign the obligation carries) rather than a derivation; automated provers are strong at the latter and weak at the former. Supplying the witness explicitly and automating the rest is therefore not a workaround but the appropriate split.

abbreviation $dos_1 \equiv \mathit{DOSUpdate}\ \mathit{assert}[a,r^c]\ []$
abbreviation $dos_2 \equiv \mathit{DOSUpdate}\ \mathit{challenge}[b,a,r^c]\ dos_1$
abbreviation $dos_3 \equiv \mathit{DOSUpdate}\ \mathit{justify}[a,n^c \vdash^{\oplus} r^c]\ dos_2$
abbreviation $dos_4 \equiv \mathit{DOSUpdate}\ \mathit{question}[c,b,r^c]\ dos_3$
abbreviation $dos_5 \equiv \mathit{DOSUpdate}\ \mathit{justify}[b,m^c \wedge^c s^c \vdash^{\ominus} r^c]\ dos_4$
abbreviation $\Gamma_1 \equiv \mathit{GammaUpdate}\ \mathit{assert}[a,r^c]\ \Gamma_B$
abbreviation $\Gamma_2 \equiv \mathit{GammaUpdate}\ \mathit{challenge}[b,a,r^c]\ \Gamma_1$
abbreviation $\Gamma_3 \equiv \mathit{GammaUpdate}\ \mathit{justify}[a,n^c \vdash^{\oplus} r^c]\ \Gamma_2$
abbreviation $\Gamma_4 \equiv \mathit{GammaUpdate}\ \mathit{question}[c,b,r^c]\ \Gamma_3$
abbreviation $\Gamma_5 \equiv \mathit{GammaUpdate}\ \mathit{justify}[b,m^c \wedge^c s^c \vdash^{\ominus} r^c]\ \Gamma_4$

lemma *B1*: *PreCond* [] Γ_B *assert*[*a*,r^c]
by (*auto simp*: Γ_B-*def intro*!: *MemberEntails*)
lemma *B2*: *PreCond* dos_1 Γ_1 *challenge*[*b*,*a*,r^c]
by (*auto simp*: Γ_B-*def GammaUpdate-def intro*!: *MemberEntails*)
lemma *B3*: *PreCond* dos_2 Γ_2 *justify*[*a*,$n^c \vdash^{\oplus} r^c$]
proof −
have Γ_2 (Done^d *challenge*[*b*,*a*,r^c]) **by** (*simp add*: *GammaUpdate-def*)
hence ∃ *j*. *j*≠*a* ∧ (Γ_2 (Done^d *question*[*j*,*a*,r^c]) ∨ Γ_2 (Done^d *challenge*[*j*,*a*,r^c]))
by (*intro exI*[*of* - *b*]) *simp*
thus *?thesis* **by** (*auto simp*: Γ_B-*def GammaUpdate-def intro*!: *MemberEntails*)
qed
lemma *B4*: *PreCond* dos_3 Γ_3 *question*[*c*,*b*,r^c]
by (*intro PreCond.simps*[*THEN iffD2*] *conjI exI*[*of* - ⊖])
(*auto simp*: Γ_B-*def GammaUpdate-def intro*!: *MemberEntails*)
lemma *B5*: *PreCond* dos_4 Γ_4 *justify*[*b*,$m^c \wedge^c s^c \vdash^{\ominus} r^c$]
proof −
have Γ_4 (Done^d *question*[*c*,*b*,r^c]) **by** (*simp add*: *GammaUpdate-def*)
hence ∃ *j*. *j*≠*b* ∧ (Γ_4 (Done^d *question*[*j*,*b*,r^c]) ∨ Γ_4 (Done^d *challenge*[*j*,*b*,r^c]))
by (*intro exI*[*of* - *c*]) *simp*

thus *?thesis* **by** (*auto simp*: Γ_B*-def GammaUpdate-def intro*!: *MemberEntails*)
qed
lemma *B6*: *PreCond dos*$_5$ Γ_5 *retract*[*a*,r^c,$\oplus$]
by (*auto simp*: Γ_B*-def GammaUpdate-def intro*!: *MemberEntails*)

theorem *BrigadeDialogue*: *FatioCheckRec Brigade* [] Γ_B
using *B1 B2 B3 B4 B5 B6* **by** *simp*

7.2 A created example: role reversal with a negative retraction

Seven locutions over two topics: after the exchange on p is settled by b RETRACTING ITS NEGATIVE obligation (exercising the $\ominus$-clause of retract), the roles reverse and b becomes the asserter of q, questioned by a.

definition Γ_C :: *BDF*$\Rightarrow$*bool* **where** $\Gamma_C \equiv \lambda x.$
$(\exists j.\ j{\neq}a \wedge x = \mathcal{D}^d\ a\ (\mathcal{B}^d\ j\ (\mathcal{B}^d\ a\ \{p^c\}^d)))$
$\vee\ (\exists k.\ k{\neq}b \wedge x = \mathcal{D}^d\ b\ (\mathcal{B}^d\ k\ (\neg^d(\mathcal{B}^d\ b\ \{p^c\}^d))))$
$\vee\ (\exists k.\ k{\neq}b \wedge x = \mathcal{D}^d\ b\ (\mathcal{B}^d\ k\ (\mathcal{D}^d\ b\ (\mathit{ExJD}\ a \oplus (p^c)))))$
$\vee\ (\exists k.\ k{\neq}a \wedge x = \mathcal{D}^d\ a\ (\mathcal{B}^d\ k\ (\mathcal{B}^d\ a\ (\mathit{EntD}\ (n^c) \oplus (p^c)))))$
$\vee\ (\exists j.\ j{\neq}b \wedge x = \mathcal{D}^d\ b\ (\mathcal{B}^d\ j\ (\neg^d\neg^d(\mathcal{B}^d\ b\ \{p^c\}^d))))$
$\vee\ (\exists j.\ j{\neq}b \wedge x = \mathcal{D}^d\ b\ (\mathcal{B}^d\ j\ (\mathcal{B}^d\ b\ \{q^c\}^d)))$
$\vee\ (\exists k.\ k{\neq}a \wedge x = \mathcal{D}^d\ a\ (\mathcal{B}^d\ k\ (\mathcal{D}^d\ a\ (\mathit{ExJD}\ b \oplus (q^c)))))$
$\vee\ (\exists k.\ k{\neq}b \wedge x = \mathcal{D}^d\ b\ (\mathcal{B}^d\ k\ (\mathcal{B}^d\ b\ (\mathit{EntD}\ (m^c) \oplus (q^c)))))$

abbreviation *Cascade* $\equiv$
[*assert*[*a*,p^c], *challenge*[*b*,*a*,p^c], *justify*[*a*,$n^c{\vdash}^{\oplus}p^c$], *retract*[*b*,p^c,$\ominus$],
assert[*b*,q^c], *question*[*a*,*b*,q^c], *justify*[*b*,$m^c{\vdash}^{\oplus}q^c$]]

abbreviation $cd_1 \equiv$ *DOSUpdate assert*[*a*,p^c] []
abbreviation $cd_2 \equiv$ *DOSUpdate challenge*[*b*,*a*,p^c] cd_1
abbreviation $cd_3 \equiv$ *DOSUpdate justify*[*a*,$n^c{\vdash}^{\oplus}p^c$] cd_2
abbreviation $cd_4 \equiv$ *DOSUpdate retract*[*b*,p^c,$\ominus$] cd_3
abbreviation $cd_5 \equiv$ *DOSUpdate assert*[*b*,q^c] cd_4
abbreviation $cd_6 \equiv$ *DOSUpdate question*[*a*,*b*,q^c] cd_5
abbreviation $\Delta_1 \equiv$ *GammaUpdate assert*[*a*,p^c] Γ_C
abbreviation $\Delta_2 \equiv$ *GammaUpdate challenge*[*b*,*a*,p^c] Δ_1
abbreviation $\Delta_3 \equiv$ *GammaUpdate justify*[*a*,$n^c{\vdash}^{\oplus}p^c$] Δ_2
abbreviation $\Delta_4 \equiv$ *GammaUpdate retract*[*b*,p^c,$\ominus$] Δ_3
abbreviation $\Delta_5 \equiv$ *GammaUpdate assert*[*b*,q^c] Δ_4
abbreviation $\Delta_6 \equiv$ *GammaUpdate question*[*a*,*b*,q^c] Δ_5

lemma *C1*: *PreCond* [] Γ_C *assert*[*a*,p^c]
by (*auto simp*: Γ_C*-def intro*!: *MemberEntails*)
lemma *C2*: *PreCond* cd_1 Δ_1 *challenge*[*b*,*a*,p^c]
by (*auto simp*: Γ_C*-def GammaUpdate-def intro*!: *MemberEntails*)
lemma *C3*: *PreCond* cd_2 Δ_2 *justify*[*a*,$n^c{\vdash}^{\oplus}p^c$]
proof −
have Δ_2 (*Done*^{d} *challenge*[*b*,*a*,p^c]) **by** (*simp add*: *GammaUpdate-def*)
hence $\exists j.\ j{\neq}a \wedge (\Delta_2\ (\mathit{Done}^d\ \mathit{question}[j,a,p^c]) \vee \Delta_2\ (\mathit{Done}^d\ \mathit{challenge}[j,a,p^c]))$

by (*intro exI*[*of* - *b*]) *simp*
thus *?thesis* **by** (*auto simp*: Γ_C-*def GammaUpdate-def intro*!: *MemberEntails*)
qed
lemma *C4*: *PreCond* cd_3 Δ_3 *retract*[*b*,p^c,$\ominus$]
by (*auto simp*: Γ_C-*def GammaUpdate-def intro*!: *MemberEntails*)
lemma *C5*: *PreCond* cd_4 Δ_4 *assert*[*b*,q^c]
by (*auto simp*: Γ_C-*def GammaUpdate-def intro*!: *MemberEntails*)
lemma *C6*: *PreCond* cd_5 Δ_5 *question*[*a*,*b*,q^c]
by (*intro PreCond.simps*[*THEN iffD2*] *conjI exI*[*of* - $\oplus$])
(*auto simp*: Γ_C-*def GammaUpdate-def intro*!: *MemberEntails*)
lemma *C7*: *PreCond* cd_6 Δ_6 *justify*[*b*,$m^c \vdash^{\oplus} q^c$]
proof −
have Δ_6 (*Done*^{d} *question*[*a*,*b*,q^c]) **by** (*simp add*: *GammaUpdate-def*)
hence $\exists j.\ j \neq b \wedge (\Delta_6\ (Done^d\ question[j,b,q^c]) \vee \Delta_6\ (Done^d\ challenge[j,b,q^c]))$
by (*intro exI*[*of* - *a*]) *simp*
thus *?thesis* **by** (*auto simp*: Γ_C-*def GammaUpdate-def intro*!: *MemberEntails*)
qed

theorem *CascadeDialogue*: *FatioCheckRec Cascade* [] Γ_C
using *C1 C2 C3 C4 C5 C6 C7* **by** *simp*

end

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